```
# Import Libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
# Set style for better graphs
sns.set(style="whitegrid")
```

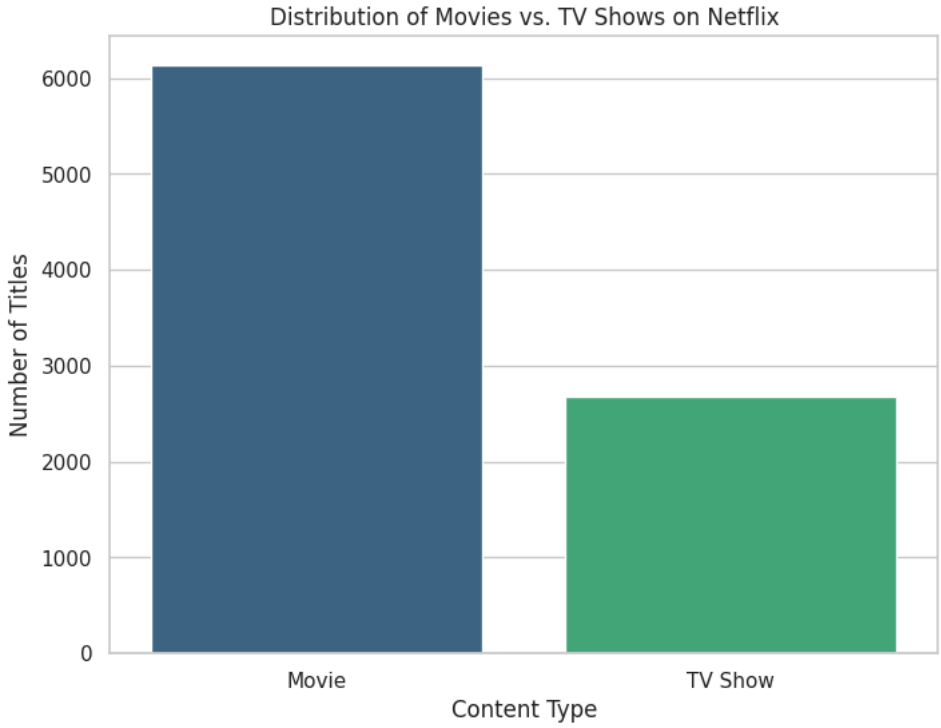
```
df = pd.read_csv('/content/netflix.csv.txt')
```

Ratio of Movies vs. TV Shows on Netflix

Based on the current dataset, the distribution of content types is as follows:

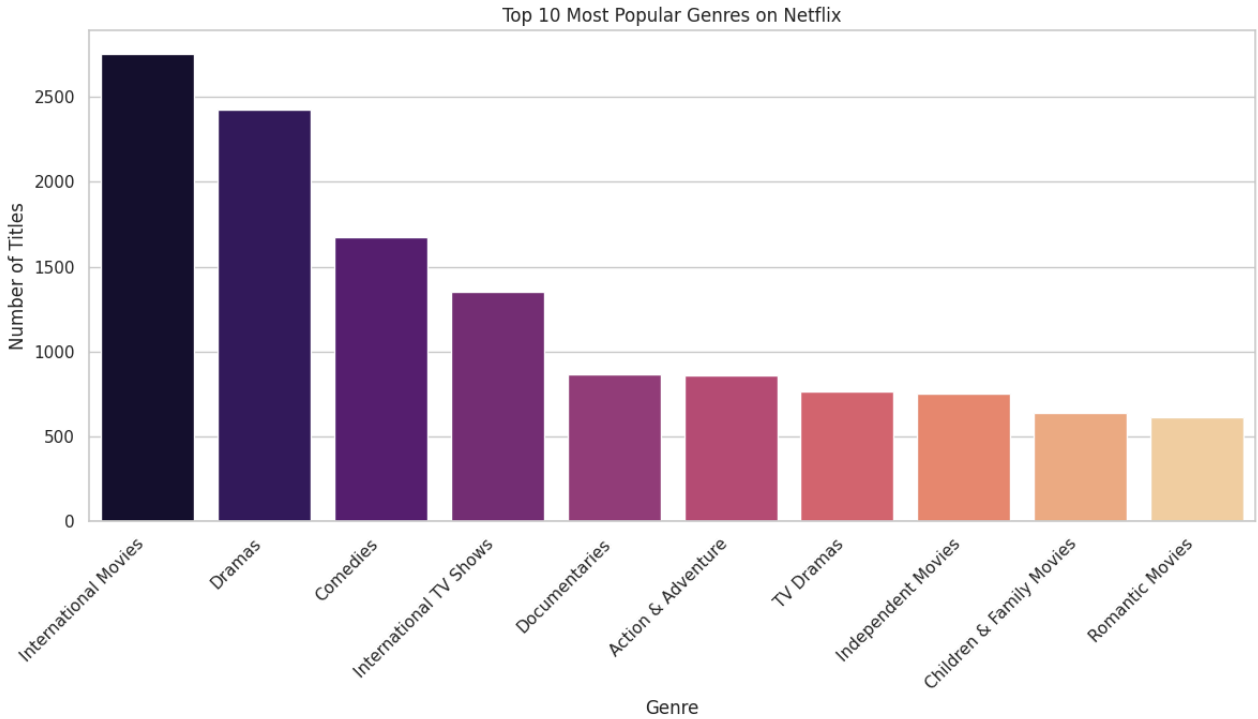
```
#Jagriti Tiwari
print(f"Distribution of Content Types:\n{content_type_counts}")
print(f"\nPercentage of Movies: {movies_percentage:.2f}%")
print(f"Percentage of TV Shows: {tv_shows_percentage:.2f}%")

# Plotting the distribution
plt.figure(figsize=(8, 6))
sns.barplot(x=content_type_counts.index, y=content_type_counts.values, hue=content_type_counts.index, palette='viridis', leg
plt.title('Distribution of Movies vs. TV Shows on Netflix')
plt.xlabel('Content Type')
plt.ylabel('Number of Titles')
plt.show()
```

```
Distribution of Content Types:
type
Movie      6131
TV Show    2676
Name: count, dtype: int64

Percentage of Movies: 69.62%
Percentage of TV Shows: 30.38%
```



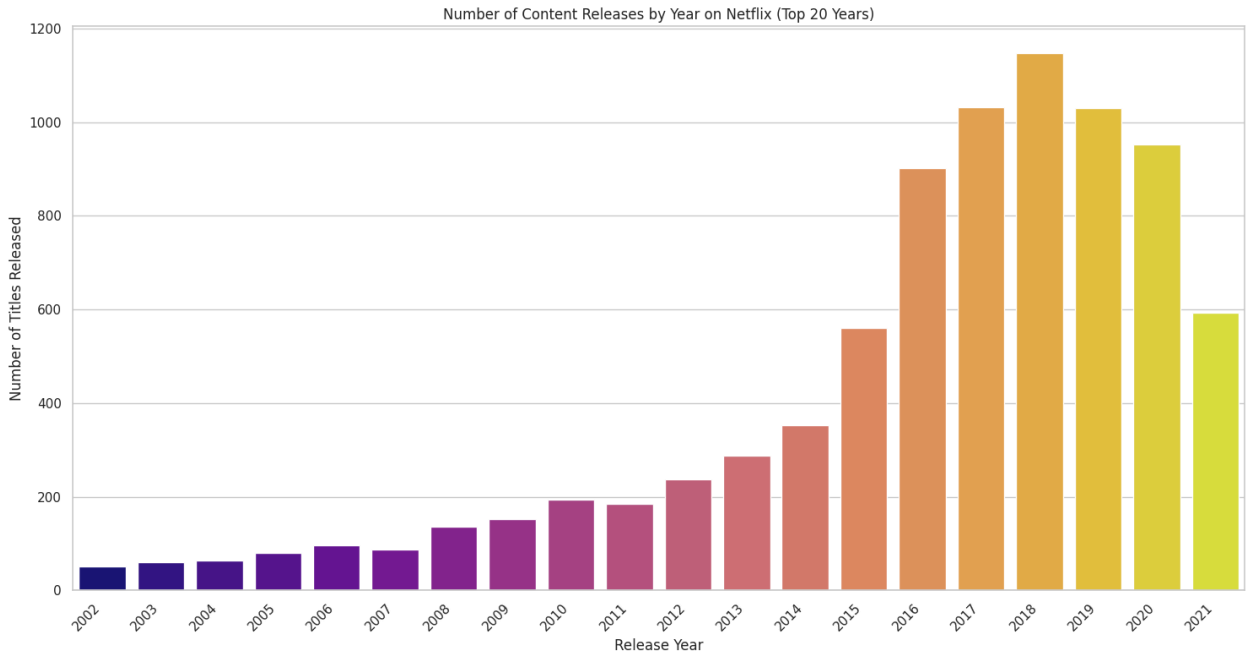

```
#Jagriti Tiwari
genres = df['listed_in'].str.split(', ', expand=True).stack()
genre_counts = genres.value_counts().head(10)

plt.figure(figsize=(12, 7))
sns.barplot(x=genre_counts.index, y=genre_counts.values, hue=genre_counts.index, palette='magma', legend=False)
plt.title('Top 10 Most Popular Genres on Netflix')
plt.xlabel('Genre')
plt.ylabel('Number of Titles')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
```

```
plt.show()

print(f"Top 10 Most Popular Genres on Netflix:\n{genre_counts}")

print("\nINSIGHTS:")
```



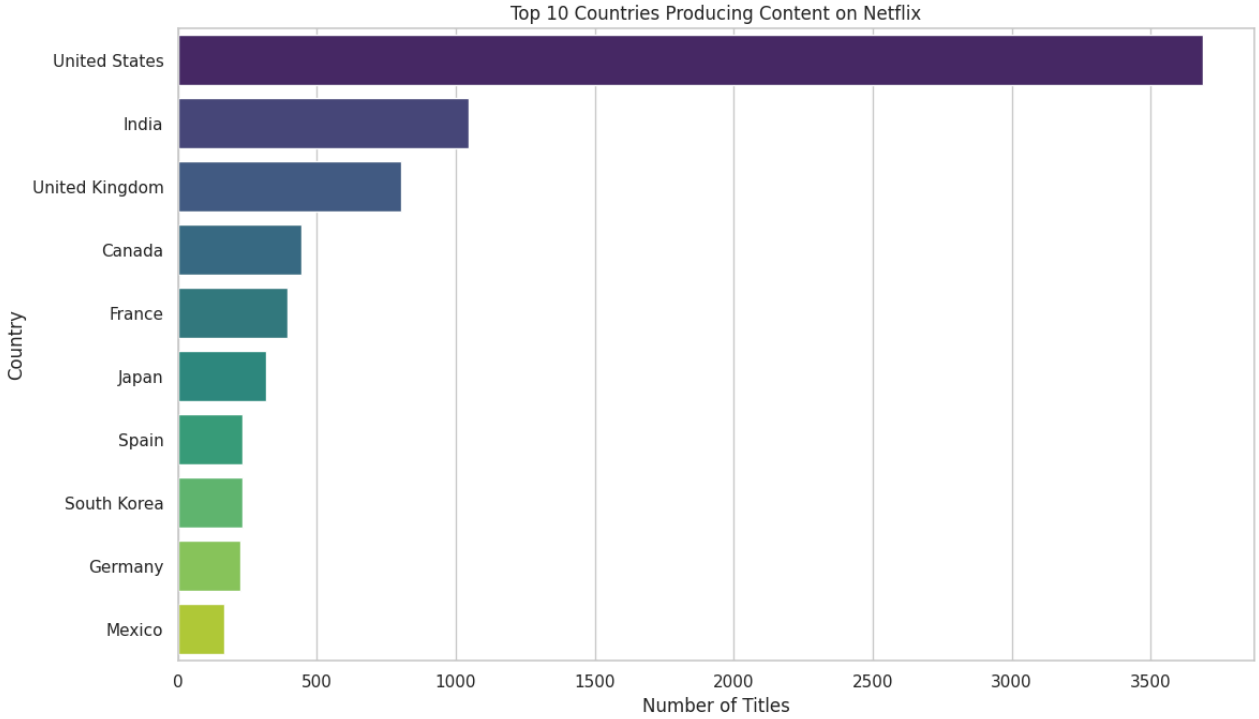

```
Top 10 Most Popular Genres on Netflix:
International Movies        2752
Dramas                      2427
Comedies                    1674
International TV Shows      1351
Documentaries                869
Action & Adventure           859
TV Dramas                    763
Independent Movies           756
Children & Family Movies     641
Romantic Movies              616
Name: count, dtype: int64

INSIGHTS:
```

```
#Jagriti Tiwari
release_year_counts = df['release_year'].value_counts().sort_index(ascending=False)

# Let's visualize the top 20 years for better clarity, or adjust as needed
plt.figure(figsize=(15, 8))
sns.barplot(x=release_year_counts.head(20).index, y=release_year_counts.head(20).values, hue=release_year_counts.head(20).i
plt.title('Number of Content Releases by Year on Netflix (Top 20 Years)')
plt.xlabel('Release Year')
plt.ylabel('Number of Titles Released')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()

print(f"Number of Content Releases by Year on Netflix (Top 20 Years):\n{release_year_counts.head(20)}")
```



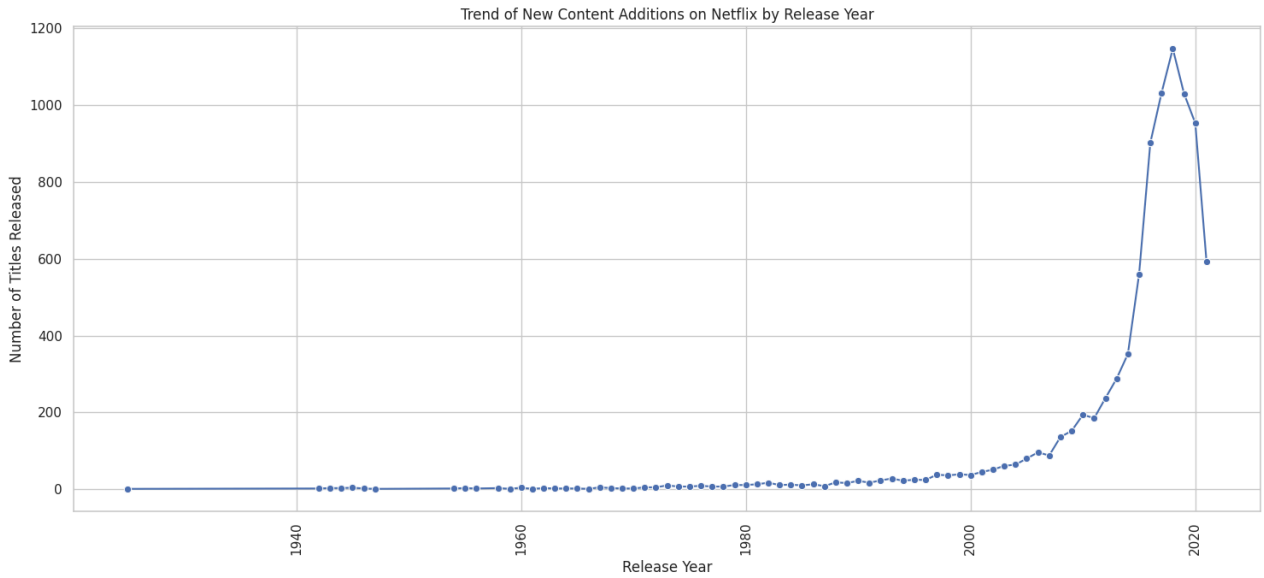

```
Number of Content Releases by Year on Netflix (Top 20 Years):
release_year
2021     592
2020     953
2019    1030
2018    1147
2017    1032
2016     902
2015     560
2014     352
2013     288
2012     237
2011     185
2010     194
2009     152
2008     136
2007      88
2006      96
2005      80
2004      64
2003      61
2002      51
Name: count, dtype: int64
```

```
#Jagriti Tiwari
countries = df['country'].str.split(', ', expand=True).stack()
country_counts = countries.value_counts().head(10)

plt.figure(figsize=(12, 7))
sns.barplot(x=country_counts.values, y=country_counts.index, hue=country_counts.index, palette='viridis', legend=False)
plt.title('Top 10 Countries Producing Content on Netflix')
plt.xlabel('Number of Titles')
plt.ylabel('Country')
plt.tight_layout()
plt.show()

print(f"Top 10 Countries Producing Content on Netflix:\n{country_counts}")
```



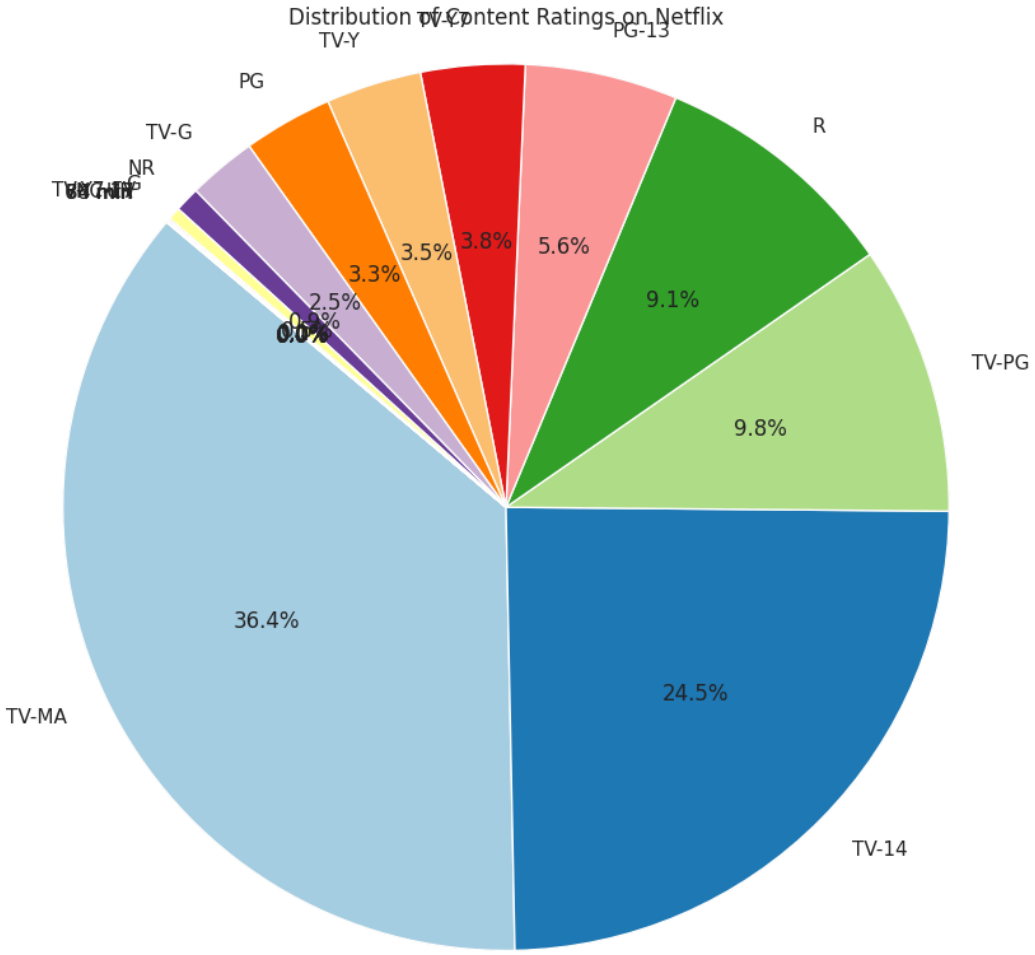

```
Top 10 Countries Producing Content on Netflix:
United States     3689
India             1046
United Kingdom     804
Canada             445
France             393
Japan              318
Spain              232
South Korea        231
Germany            226
Mexico             169
Name: count, dtype: int64
```

```
#Jagriti Tiwari
plt.figure(figsize=(15, 7))
sns.lineplot(x=release_year_counts.index, y=release_year_counts.values, marker='o')
plt.title('Trend of New Content Additions on Netflix by Release Year')
plt.xlabel('Release Year')
plt.ylabel('Number of Titles Released')
plt.xticks(rotation=90)
plt.grid(True)
plt.tight_layout()
plt.show()

print(f"Trend of New Content Additions on Netflix by Release Year:\n{release_year_counts}")
```



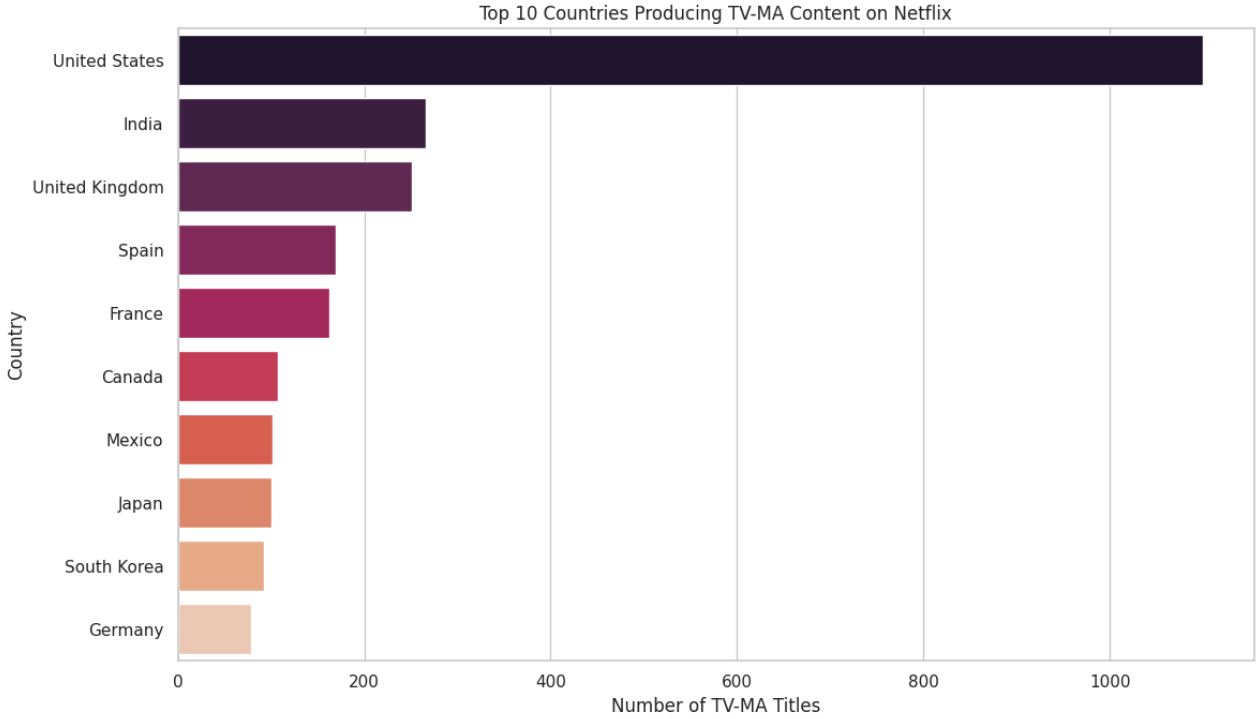

```
Trend of New Content Additions on Netflix by Release Year:
release_year
2021     592
2020     953
2019    1030
2018    1147
2017    1032
        ...
1945       4
1944       3
1943       3
1942       2
1925       1
Name: count, Length: 74, dtype: int64
```

```
#Jagriti Tiwari
rating_counts = df['rating'].value_counts()

plt.figure(figsize=(10, 8))
plt.pie(rating_counts, labels=rating_counts.index, autopct='%1.1f%%', startangle=140, colors=plt.cm.Paired.colors)
plt.title('Distribution of Content Ratings on Netflix')
plt.axis('equal') # Equal aspect ratio ensures that pie is drawn as a circle.
plt.tight_layout()
plt.show()

print(f"Distribution of Content Ratings on Netflix:\n{rating_counts}")
```



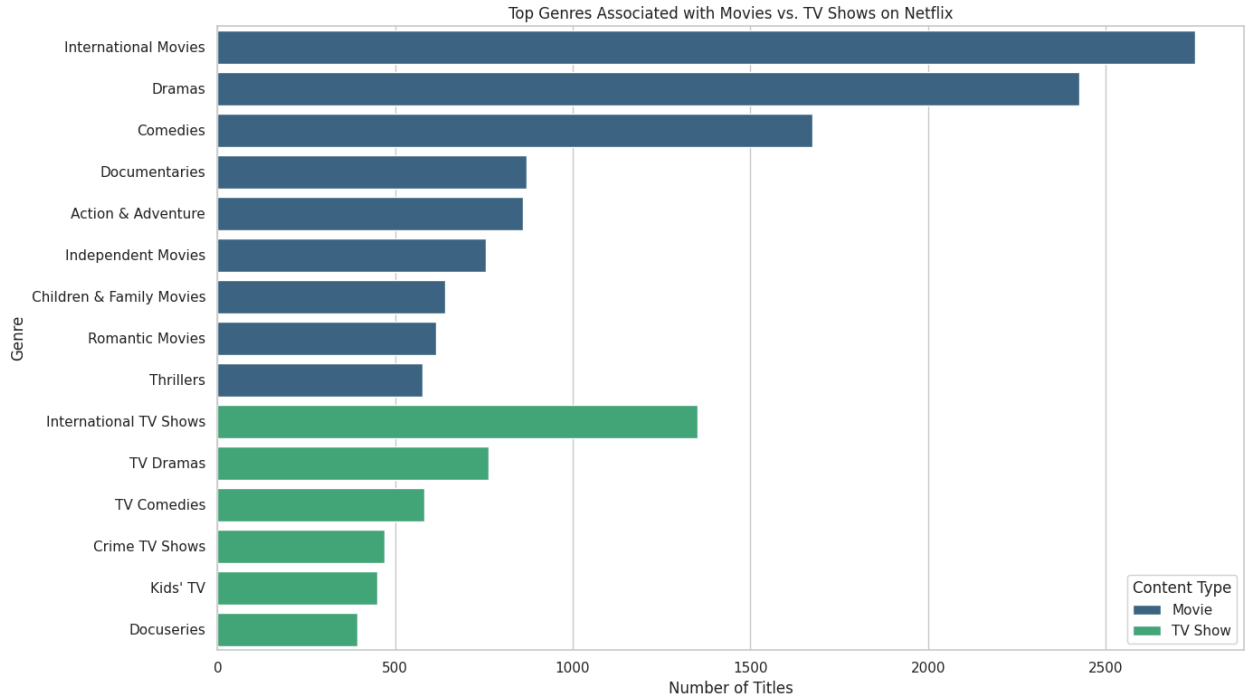

```
Distribution of Content Ratings on Netflix:
rating
TV-MA       3207
TV-14       2160
TV-PG        863
R            799
PG-13        490
TV-Y7        334
TV-Y         307
PG           287
TV-G         220
NR            80
G             41
TV-Y7-FV       6
NC-17          3
UR             3
74 min         1
84 min         1
66 min         1
Name: count, dtype: int64
```

```
#Jagriti tiwari
tv_ma_content = df[df['rating'] == 'TV-MA']
tv_ma_countries = tv_ma_content['country'].str.split(', ', expand=True).stack()
tv_ma_country_counts = tv_ma_countries.value_counts().head(10)

plt.figure(figsize=(12, 7))
sns.barplot(x=tv_ma_country_counts.values, y=tv_ma_country_counts.index, hue=tv_ma_country_counts.index, palette='rocket', l
plt.title('Top 10 Countries Producing TV-MA Content on Netflix')
plt.xlabel('Number of TV-MA Titles')
plt.ylabel('Country')
plt.tight_layout()
plt.show()

print(f"Top 10 Countries Producing TV-MA Content on Netflix:\n{tv_ma_country_counts}")
```




```
Top 10 Countries Producing TV-MA Content on Netflix:
United States     1100
India              266
United Kingdom     251
Spain              170
France             163
Canada             107
Mexico             102
Japan              101
South Korea         92
Germany             79
Name: count, dtype: int64
```

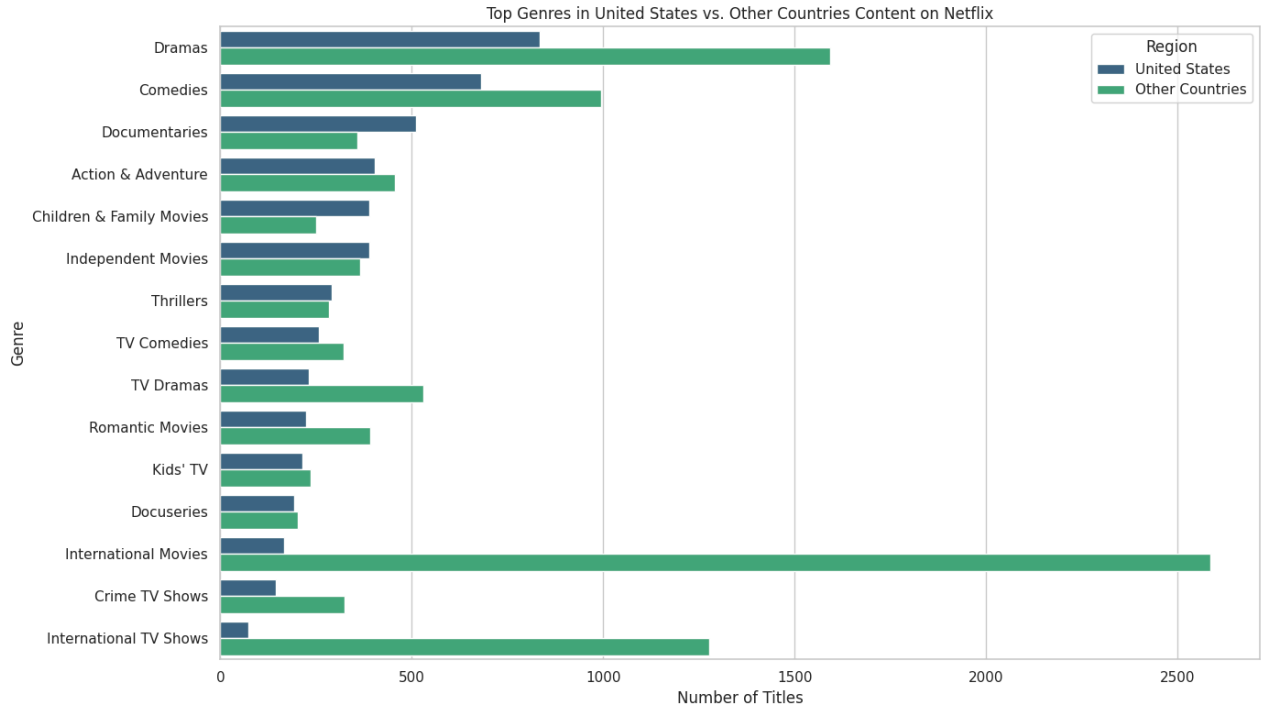
```
#Jagriti tiwari
movies_df = df[df['type'] == 'Movie']
tv_shows_df = df[df['type'] == 'TV Show']

movie_genres = movies_df['listed_in'].str.split(', ', expand=True).stack()
movie_genre_counts = movie_genres.value_counts().reset_index()
movie_genre_counts.columns = ['genre', 'count']
movie_genre_counts['type'] = 'Movie'

tv_show_genres = tv_shows_df['listed_in'].str.split(', ', expand=True).stack()
tv_show_genre_counts = tv_show_genres.value_counts().reset_index()
tv_show_genre_counts.columns = ['genre', 'count']
tv_show_genre_counts['type'] = 'TV Show'

combined_genre_counts = pd.concat([movie_genre_counts, tv_show_genre_counts])

# Get the top N overall genres to display, ensuring representation from both types
top_overall_genres = combined_genre_counts.groupby('genre')['count'].sum().nlargest(15).index
filtered_combined_genre_counts = combined_genre_counts[combined_genre_counts['genre'].isin(top_overall_genres)]

plt.figure(figsize=(14, 8))
sns.barplot(data=filtered_combined_genre_counts, x='count', y='genre', hue='type', palette='viridis')
plt.title('Top Genres Associated with Movies vs. TV Shows on Netflix')
plt.xlabel('Number of Titles')
plt.ylabel('Genre')
plt.legend(title='Content Type')
plt.tight_layout()
plt.show()

print(f"Top Genres Associated with Movies vs. TV Shows on Netflix:\n{combined_genre_counts}")
```




```
Top Genres Associated with Movies vs. TV Shows on Netflix:
                         genre  count     type
0           International Movies   2752    Movie
1                         Dramas   2427    Movie
2                       Comedies   1674    Movie
3                  Documentaries    869    Movie
4             Action & Adventure    859    Movie
5             Independent Movies    756    Movie
6       Children & Family Movies    641    Movie
7                Romantic Movies    616    Movie
8                       Thrillers   577    Movie
9               Music & Musicals    375    Movie
10                Horror Movies    357    Movie
11               Stand-Up Comedy   343    Movie
12              Sci-Fi & Fantasy   243    Movie
13                Sports Movies    219    Movie
14               Classic Movies    116    Movie
15                 LGBTQ Movies    102    Movie
16                  Cult Movies     71    Movie
17                Anime Features    71    Movie
18          Faith & Spirituality    65    Movie
19                        Movies    57    Movie
0         International TV Shows   1351  TV Show
1                      TV Dramas   763  TV Show
2                    TV Comedies   581  TV Show
3                 Crime TV Shows   470  TV Show
4                       Kids' TV   451  TV Show
5                     Docuseries   395  TV Show
6              Romantic TV Shows   370  TV Show
7                     Reality TV   255  TV Show
8               British TV Shows   253  TV Show
9                    Anime Series   176  TV Show
10     Spanish-Language TV Shows   174  TV Show
11         TV Action & Adventure   168  TV Show
12                Korean TV Shows   151  TV Show
13                  TV Mysteries    98  TV Show
14           Science & Nature TV    92  TV Show
15           TV Sci-Fi & Fantasy    84  TV Show
16                      TV Horror   75  TV Show
17                 Teen TV Shows    69  TV Show
18                  TV Thrillers    57  TV Show
19  Stand-Up Comedy & Talk Shows    56  TV Show
20             Classic & Cult TV    28  TV Show
21                       TV Shows   16  TV Show
```

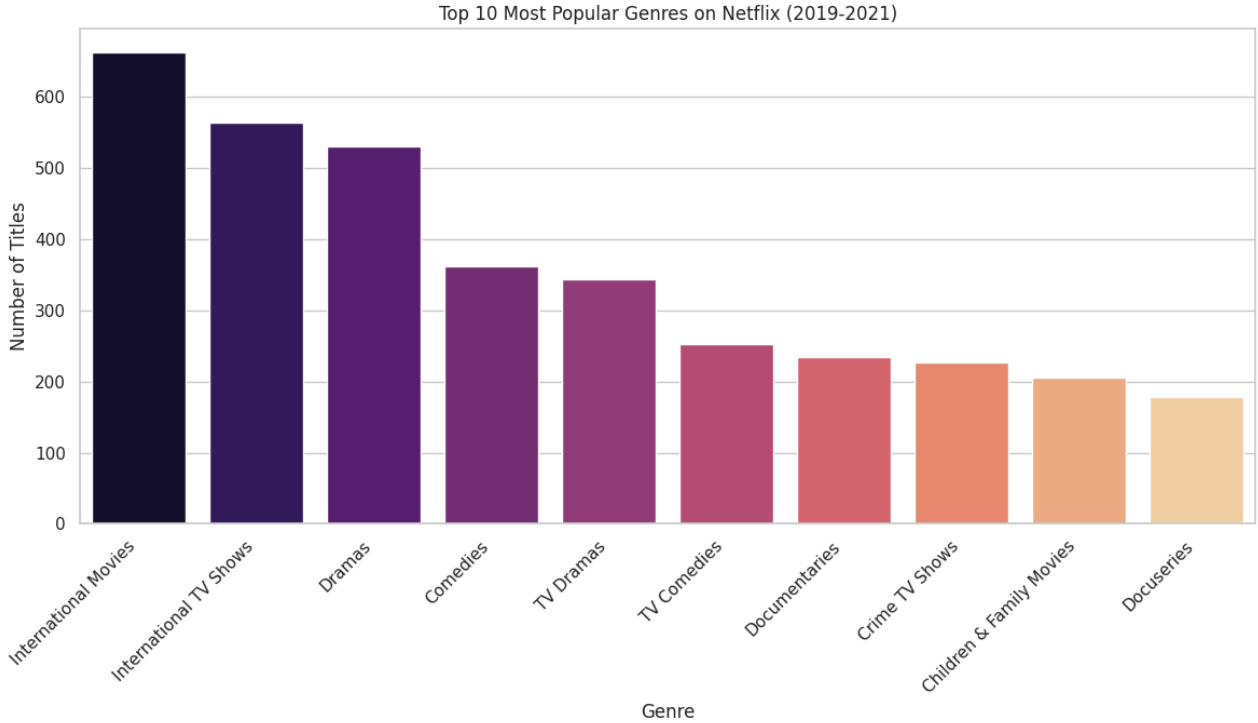
```
#Jagriti Tiwari
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Filter content involving the United States
us_only_content = df[df['country'].fillna('').str.contains('United States')]
```

```
# Filter content NOT involving the United States
non_us_content = df[~df['country'].fillna('').str.contains('United States')]

# Extract and count genres for U.S. content
us_genres = us_only_content['listed_in'].str.split(', ', expand=True).stack()
us_genre_counts = us_genres.value_counts().reset_index()
us_genre_counts.columns = ['genre', 'count']
us_genre_counts['region'] = 'United States'

# Extract and count genres for Other Countries content
other_countries_genres = non_us_content['listed_in'].str.split(', ', expand=True).stack()
other_countries_genre_counts = other_countries_genres.value_counts().reset_index()
other_countries_genre_counts.columns = ['genre', 'count']
other_countries_genre_counts['region'] = 'Other Countries'

# Combine the genre counts from both regions
combined_country_genre_counts_raw = pd.concat([us_genre_counts, other_countries_genre_counts])

# Get the top 15 overall genres across both regions for consistent comparison
top_overall_genres_country = combined_country_genre_counts_raw.groupby('genre')['count'].sum().nlargest(15).index
filtered_combined_country_genre_counts = combined_country_genre_counts_raw[combined_country_genre_counts_raw['genre'].isin(

# Plotting
plt.figure(figsize=(14, 8))
sns.barplot(data=filtered_combined_country_genre_counts, x='count', y='genre', hue='region', palette='viridis')
plt.title('Top Genres in United States vs. Other Countries Content on Netflix')
plt.xlabel('Number of Titles')
plt.ylabel('Genre')
plt.legend(title='Region')
plt.tight_layout()
plt.show()

print(f"Top Genres in United States vs. Other Countries Content on Netflix:\n{combined_country_genre_counts_raw}")
```



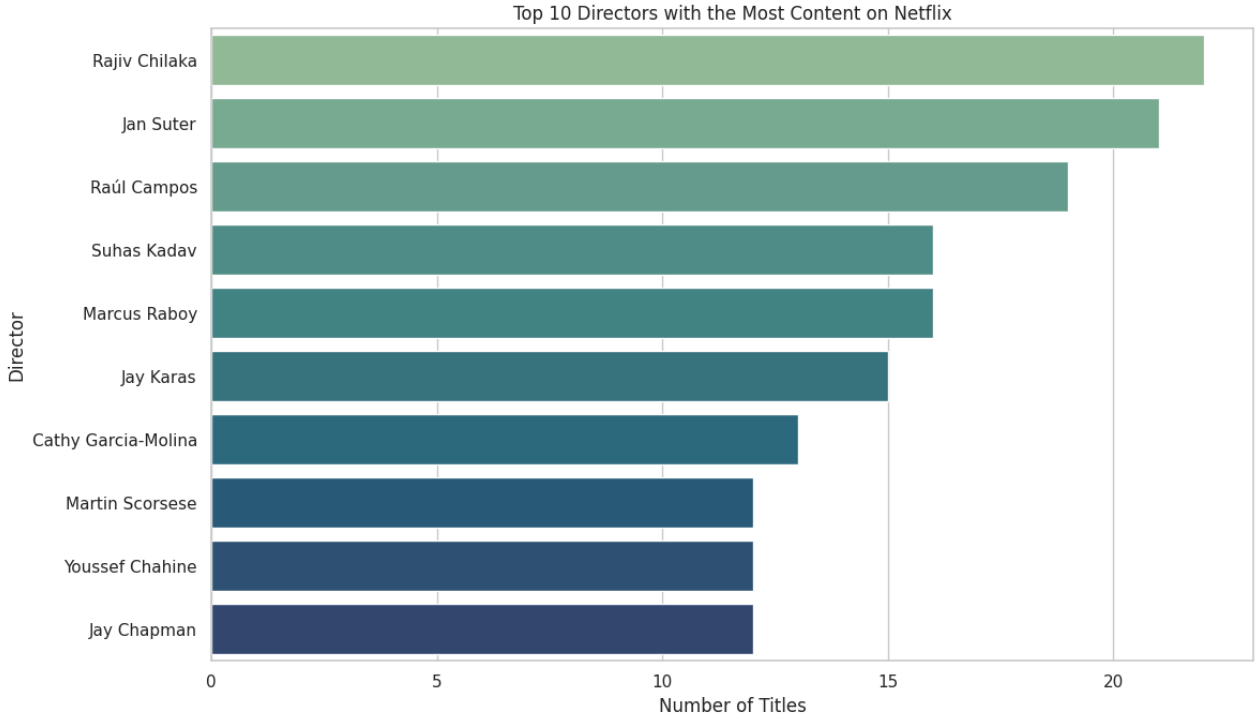

```
Top Genres in United States vs. Other Countries Content on Netflix:
                           genre  count           region
0                         Dramas    835    United States
1                       Comedies    680    United States
2                  Documentaries    512    United States
3              Action & Adventure    404    United States
4        Children & Family Movies    390    United States
..                            ...    ...              ...
37          Faith & Spirituality     23  Other Countries
38  Stand-Up Comedy & Talk Shows     23  Other Countries
39                    Cult Movies     19  Other Countries
40                       TV Shows     12  Other Countries
41              Classic & Cult TV     11  Other Countries

[84 rows x 3 columns]
```

```
#Jagriti Tiwari
# Get the most recent year in the dataset to define 'last 3 years'
latest_year = df['release_year'].max()
last_three_years = range(latest_year - 2, latest_year + 1) # Including latest_year, and two years prior

# Filter the DataFrame for content released in the last 3 years
df_last_three_years = df[df['release_year'].isin(last_three_years)]

# Extract and count genres for this filtered data
genres_last_three_years = df_last_three_years['listed_in'].str.split(', ', expand=True).stack()
genre_counts_last_three_years = genres_last_three_years.value_counts().head(10)

# Plotting the top 10 most popular genres in the last 3 years
plt.figure(figsize=(12, 7))
sns.barplot(x=genre_counts_last_three_years.index, y=genre_counts_last_three_years.values, hue=genre_counts_last_three_year
plt.title(f'Top 10 Most Popular Genres on Netflix ({latest_year - 2}-{latest_year})')
plt.xlabel('Genre')
plt.ylabel('Number of Titles')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()

print(f"Top 10 Most Popular Genres on Netflix ({latest_year - 2}-{latest_year}):\n{genre_counts_last_three_years}")
```



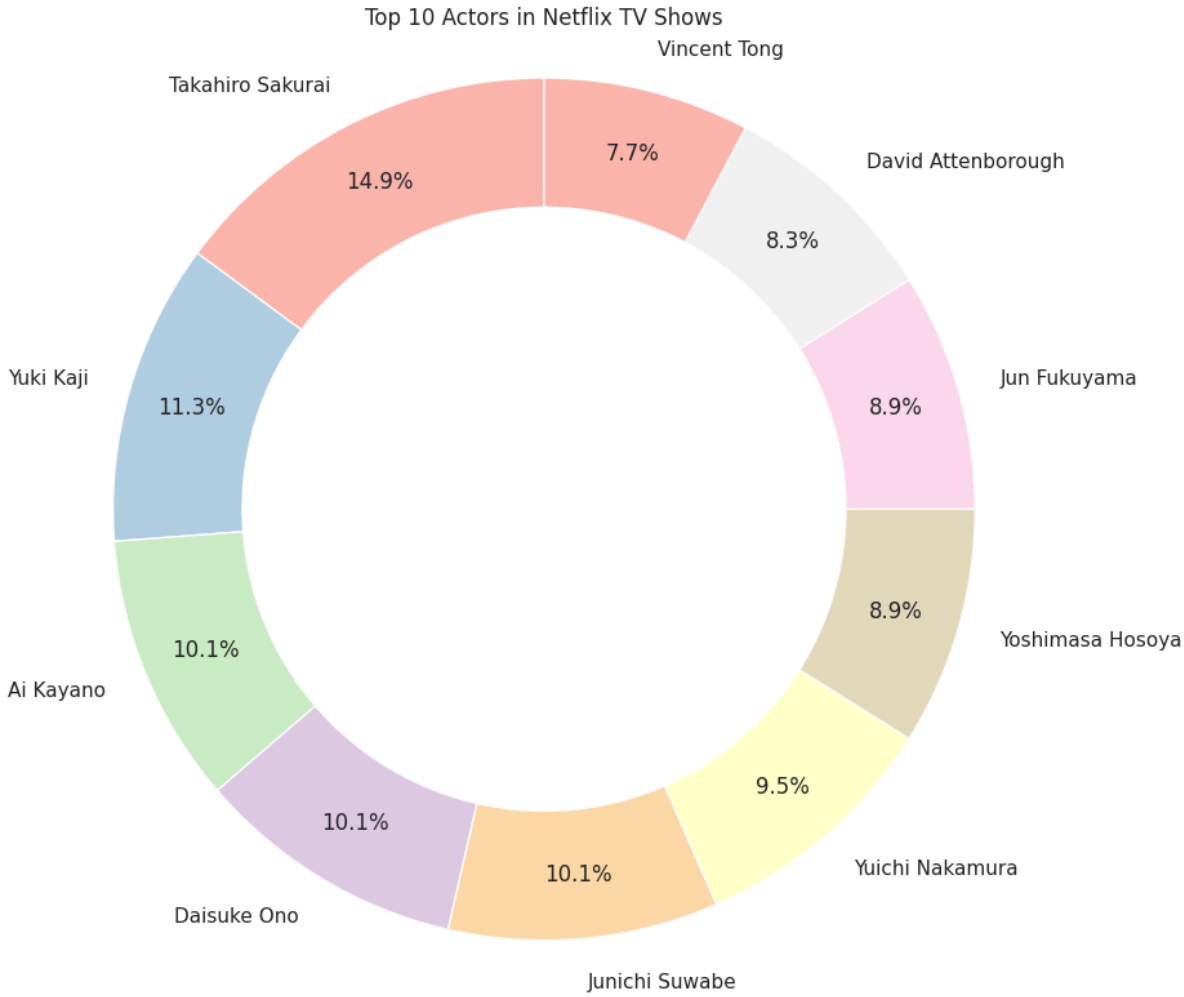

```
Top 10 Most Popular Genres on Netflix (2019-2021):
International Movies        662
International TV Shows      564
Dramas                      530
Comedies                    362
TV Dramas                   343
TV Comedies                 252
Documentaries               234
Crime TV Shows              226
Children & Family Movies    205
Docuseries                  178
Name: count, dtype: int64
```

```
#Jagriti Tiwari
# Drop rows where 'director' is NaN as they cannot be attributed to a director
directors_df = df.dropna(subset=['director']).copy()

# Split directors string into individual names and stack them to count each director individually
directors = directors_df['director'].str.split(', ', expand=True).stack()

# Count the occurrences of each director
director_counts = directors.value_counts().head(10)

# Plotting the top 10 directors
plt.figure(figsize=(12, 7))
sns.barplot(x=director_counts.values, y=director_counts.index, hue=director_counts.index, palette='crest', legend=False)
plt.title('Top 10 Directors with the Most Content on Netflix')
plt.xlabel('Number of Titles')
plt.ylabel('Director')
plt.tight_layout()
plt.show()

print(f"Top 10 Directors with the Most Content on Netflix:\n{director_counts}")
```




```
Top 10 Directors with the Most Content on Netflix:
Rajiv Chilaka          22
Jan Suter              21
Raúl Campos            19
Suhas Kadav            16
Marcus Raboy           16
Jay Karas              15
Cathy Garcia-Molina    13
Martin Scorsese        12
Youssef Chahine        12
Jay Chapman            12
Name: count, dtype: int64
```

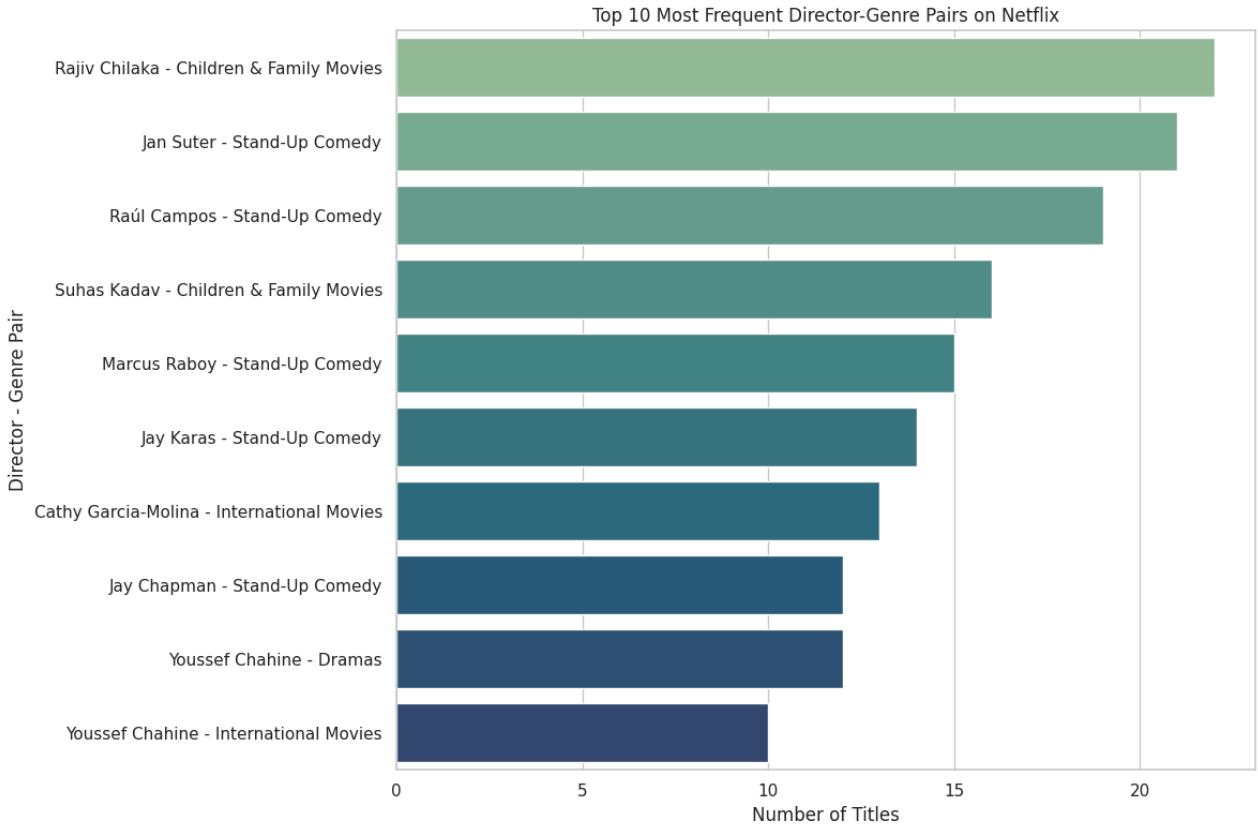
```
#Jagriti Tiwari
# Filter for TV shows and drop rows where 'cast' is NaN
tv_show_cast_df = df[df['type'] == 'TV Show'].dropna(subset=['cast']).copy()

# Split the 'cast' string into individual actors and stack them
tv_show_actors = tv_show_cast_df['cast'].str.split(', ', expand=True).stack()

# Count the occurrences of each actor
actor_counts_tv_shows = tv_show_actors.value_counts().head(10)

# Plotting the donut chart
plt.figure(figsize=(10, 8))

# Create a pie chart
plt.pie(actor_counts_tv_shows, labels=actor_counts_tv_shows.index, autopct='%1.1f%%', startangle=90, pctdistance=0.85, colo

# Draw a circle in the middle to make it a donut chart
centre_circle = plt.Circle((0,0), 0.70, fc='white')
fig = plt.gcf()
fig.gca().add_artist(centre_circle)

plt.title('Top 10 Actors in Netflix TV Shows')
plt.axis('equal') # Equal aspect ratio ensures that pie is drawn as a circle.
plt.tight_layout()
plt.show()

print(f"Top 10 Actors in Netflix TV Shows:\n{actor_counts_tv_shows}")
```




```
Top 10 Actors in Netflix TV Shows:
Takahiro Sakurai      25
Yuki Kaji             19
Ai Kayano             17
Daisuke Ono           17
Junichi Suwabe        17
Yuichi Nakamura       16
Yoshimasa Hosoya      15
Jun Fukuyama          15
David Attenborough    14
Vincent Tong          13
Name: count, dtype: int64
```

```
#Jagriti Tiwari
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import itertools

# Drop rows where either 'director' or 'listed_in' is missing, as these cannot form pairs
df_filtered = df.dropna(subset=['director', 'listed_in']).copy()

# Prepare a list to store all director-genre pairs
director_genre_pairs = []

# Iterate through the filtered DataFrame to generate director-genre combinations
for index, row in df_filtered.iterrows():
    # Split directors string into individual names and strip whitespace
    directors = [d.strip() for d in row['director'].split(',')]
    # Split listed_in string into individual genres and strip whitespace
    genres = [g.strip() for g in row['listed_in'].split(',')]

    # Generate all combinations of director and genre for the current title
    for director, genre in itertools.product(directors, genres):
        director_genre_pairs.append((director, genre))

# Convert the list of pairs to a pandas Series and count frequencies
director_genre_counts = pd.Series(director_genre_pairs).value_counts().head(10)

# Prepare data for plotting by formatting the pairs as strings
plot_data = pd.DataFrame({
```

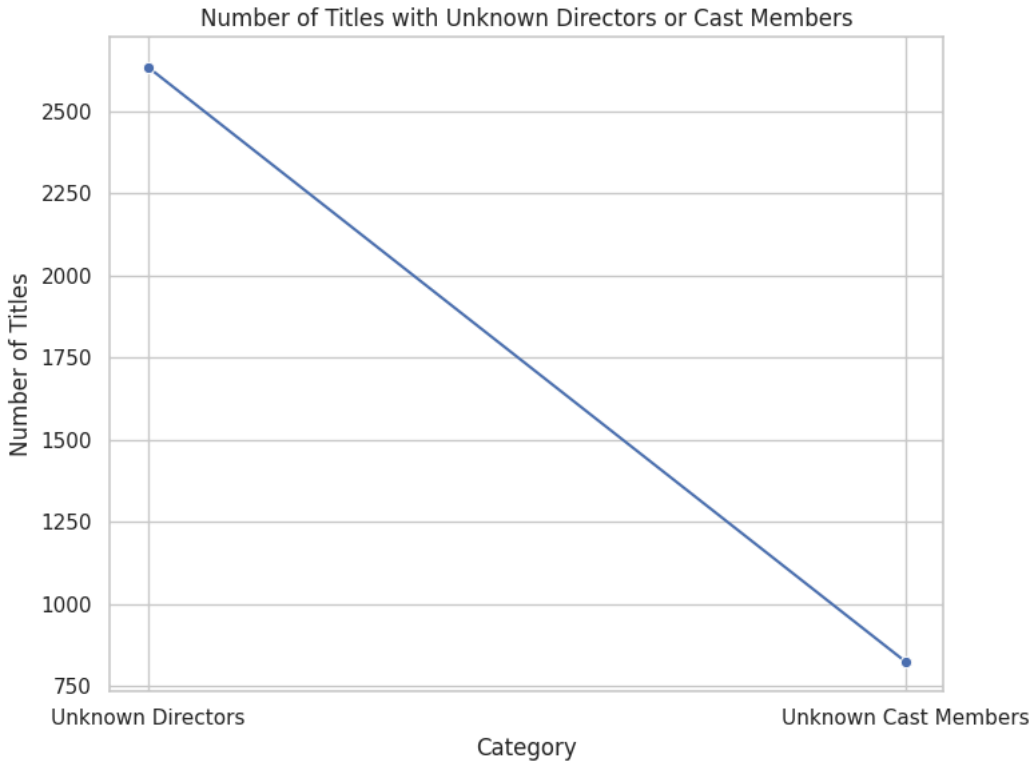
```
    'Pair': director_genre_counts.index.map(lambda x: f"{x[0]} - {x[1]}"),
    'Count': director_genre_counts.values
})

# Plotting the top 10 most frequent director-genre pairs
plt.figure(figsize=(12, 8))
sns.barplot(x='Count', y='Pair', data=plot_data, hue='Pair', palette='crest', legend=False)
plt.title('Top 10 Most Frequent Director-Genre Pairs on Netflix')
plt.xlabel('Number of Titles')
plt.ylabel('Director - Genre Pair')
plt.tight_layout()
plt.show()

print(f"Top 10 Most Frequent Director-Genre Pairs on Netflix:\n{director_genre_counts}")
```




```
Top 10 Most Frequent Director-Genre Pairs on Netflix:
(Rajiv Chilaka, Children & Family Movies)      22
(Jan Suter, Stand-Up Comedy)                   21
(Raúl Campos, Stand-Up Comedy)                 19
(Suhas Kadav, Children & Family Movies)        16
(Marcus Raboy, Stand-Up Comedy)                15
(Jay Karas, Stand-Up Comedy)                   14
(Cathy Garcia-Molina, International Movies)    13
(Jay Chapman, Stand-Up Comedy)                 12
(Youssef Chahine, Dramas)                      12
(Youssef Chahine, International Movies)        10
Name: count, dtype: int64
```

```
#Jagriti Tiwari
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Count unknown directors and cast members
unknown_directors = df['director'].isnull().sum()
unknown_cast = df['cast'].isnull().sum()

# Create a DataFrame for plotting
data_for_plot = pd.DataFrame({
    'Category': ['Unknown Directors', 'Unknown Cast Members'],
    'Count': [unknown_directors, unknown_cast]
})
```

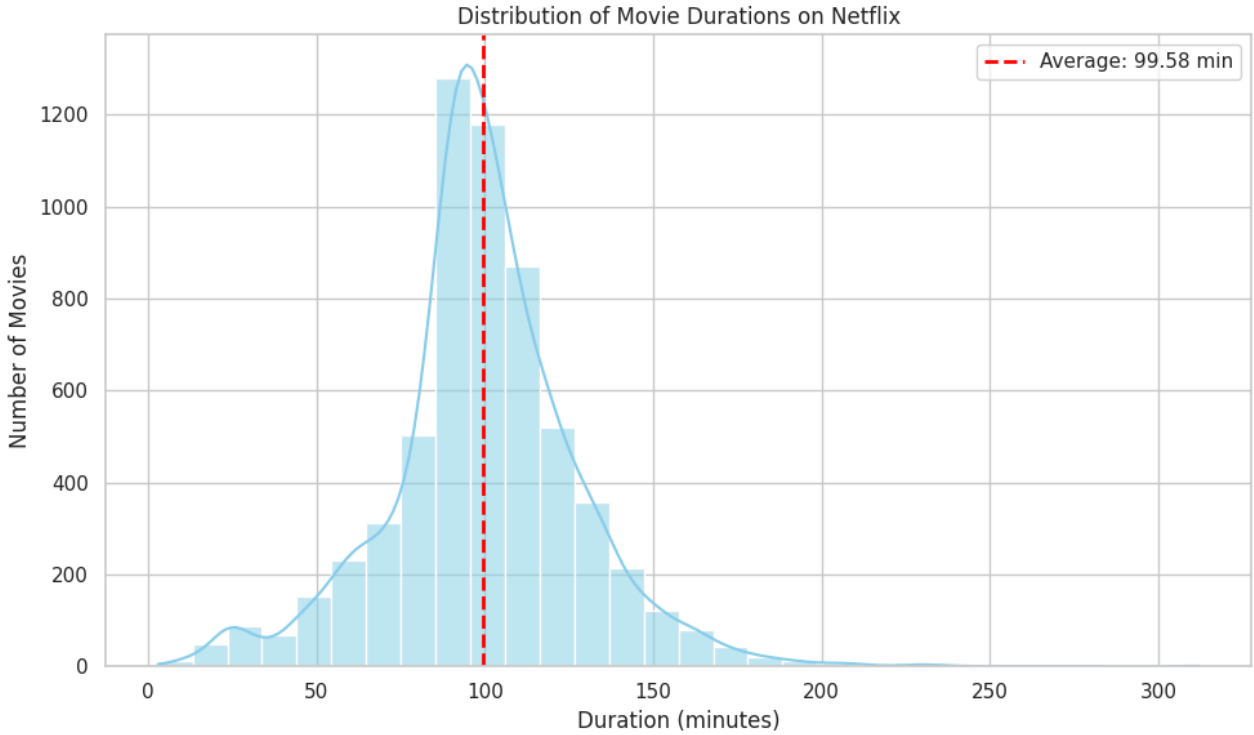
```
# Plotting the line chart
plt.figure(figsize=(8, 6))
sns.lineplot(x='Category', y='Count', data=data_for_plot, marker='o', sort=False)
plt.title('Number of Titles with Unknown Directors or Cast Members')
plt.xlabel('Category')
plt.ylabel('Number of Titles')
plt.grid(True)
plt.tight_layout()
plt.show()

print(f"Number of titles with unknown directors: {unknown_directors}")
print(f"Number of titles with unknown cast members: {unknown_cast}")
```




```
Number of titles with unknown directors: 2634
Number of titles with unknown cast members: 825
```

```
#Jagriti Tiwari
# Filter for movies only
movies_df = df[df['type'] == 'Movie'].copy()

# Extract duration in minutes, handling missing values and ensuring it's numeric
movies_df['duration_minutes'] = movies_df['duration'].str.extract(r'(\d+)\smin').astype(float)

# Calculate the average duration for movies
average_movie_duration = movies_df['duration_minutes'].mean()

print(f"Average duration of Movies on Netflix: {average_movie_duration:.2f} minutes")

# Plotting a histogram of movie durations
plt.figure(figsize=(10, 6))
sns.histplot(movies_df['duration_minutes'].dropna(), bins=30, kde=True, color='skyblue')
plt.title('Distribution of Movie Durations on Netflix')
plt.xlabel('Duration (minutes)')
plt.ylabel('Number of Movies')
plt.axvline(average_movie_duration, color='red', linestyle='dashed', linewidth=2, label=f'Average: {average_movie_duration:
plt.legend()
plt.tight_layout()
plt.show()

print(f"Average duration of Movies on Netflix: {average_movie_duration:.2f} minutes")
```

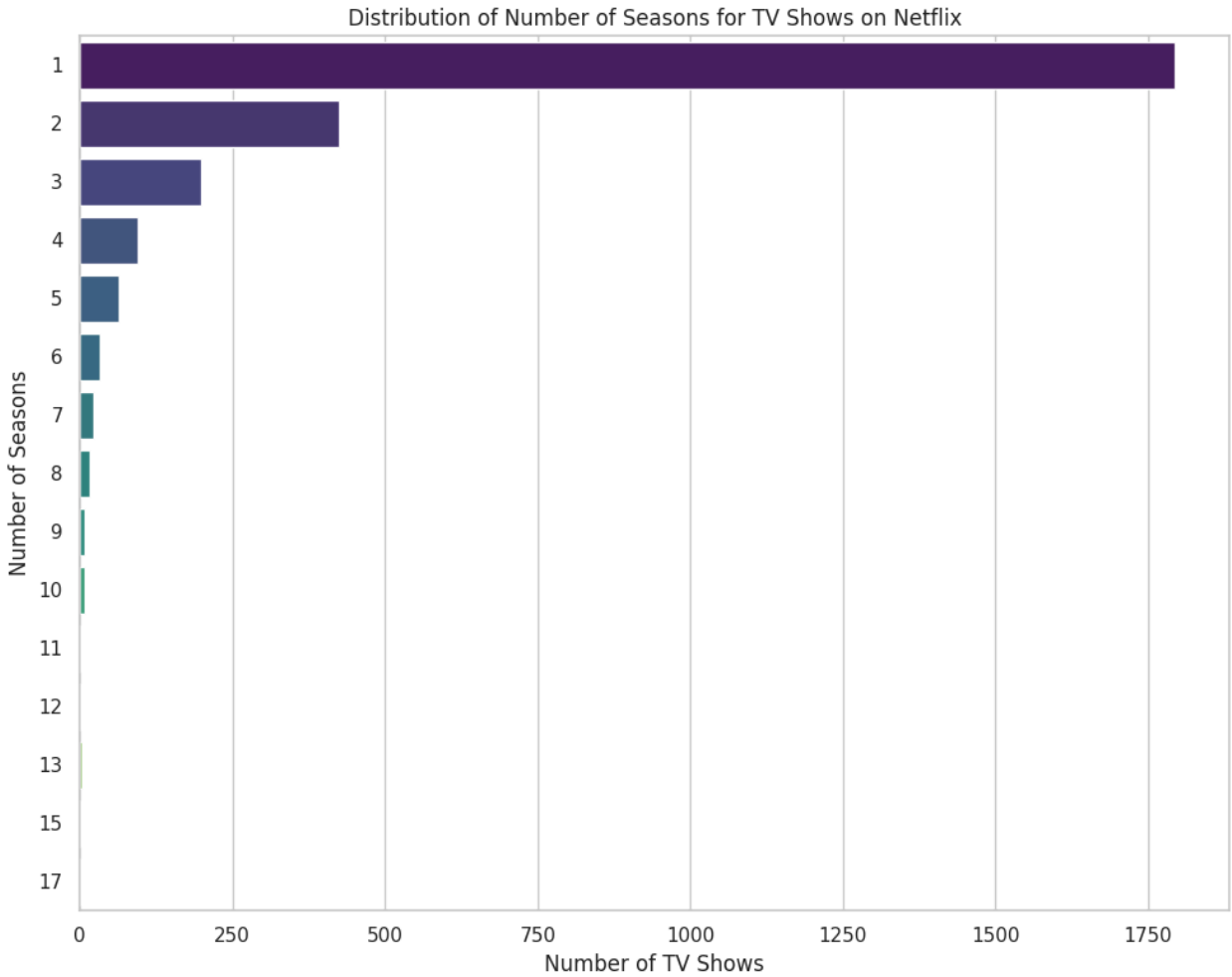
```
Average duration of Movies on Netflix: 99.58 minutes
```




```
Average duration of Movies on Netflix: 99.58 minutes
```

```
#Jagriti Tiwari
# Filter for TV Shows only
tv_shows_df = df[df['type'] == 'TV Show'].copy()

# Extract the number of seasons from the 'duration' column
# The duration column contains strings like '1 Season', '2 Seasons', etc.
# We use regex to extract the digit(s) at the beginning of the string.
tv_shows_df['num_seasons'] = tv_shows_df['duration'].str.extract(r'(\d+) Season').astype(int)

# Count the occurrences of each number of seasons
seasons_counts = tv_shows_df['num_seasons'].value_counts().sort_index()

# Get the most common number of seasons
most_common_seasons = seasons_counts.idxmax()
max_count = seasons_counts.max()

print(f"The most common number of seasons for TV shows on Netflix is: {most_common_seasons} season(s), with {max_count} oc

# Plotting the distribution of seasons with a horizontal bar graph
plt.figure(figsize=(10, 8))
sns.barplot(x=seasons_counts.values, y=seasons_counts.index.astype(str), hue=seasons_counts.index.astype(str), palette='vi
plt.title('Distribution of Number of Seasons for TV Shows on Netflix')
plt.xlabel('Number of TV Shows')
plt.ylabel('Number of Seasons')
plt.tight_layout()
plt.show()

print(f"Distribution of Number of Seasons for TV Shows on Netflix:\n{seasons_counts}")
```

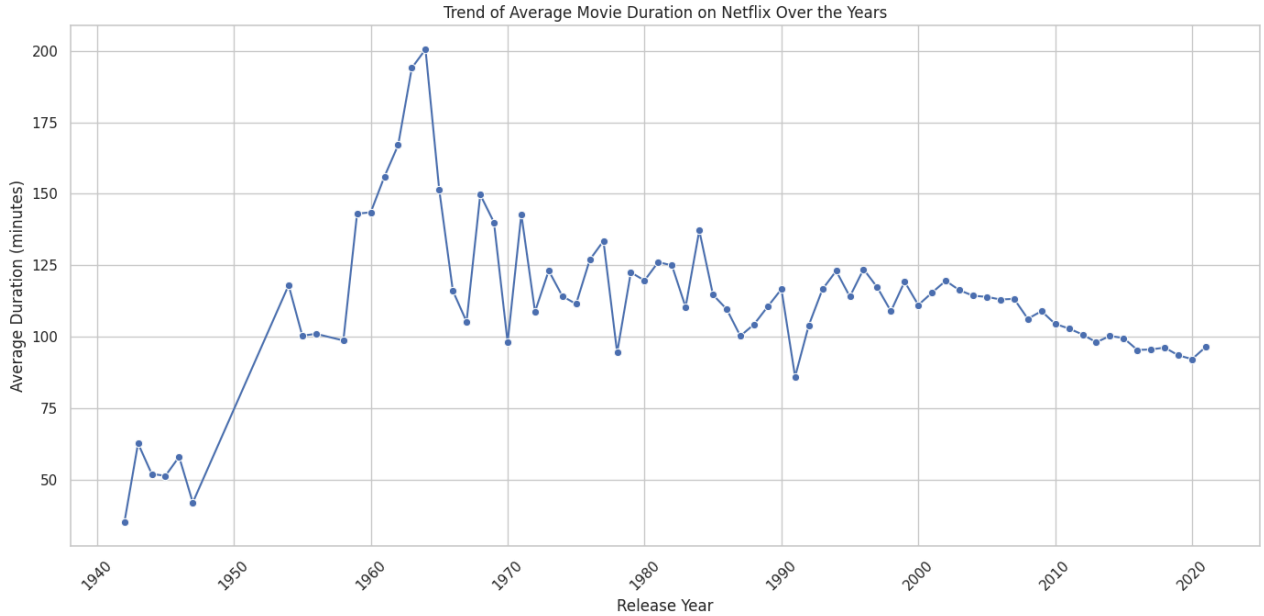
```
The most common number of seasons for TV shows on Netflix is: 1 season(s), with 1793 occurrences.
```




```
Distribution of Number of Seasons for TV Shows on Netflix:
num_seasons
1     1793
2      425
3      199
4       95
5       65
6       33
7       23
8       17
9        9
10       7
11       2
12       2
13       3
15       2
17       1
Name: count, dtype: int64
```

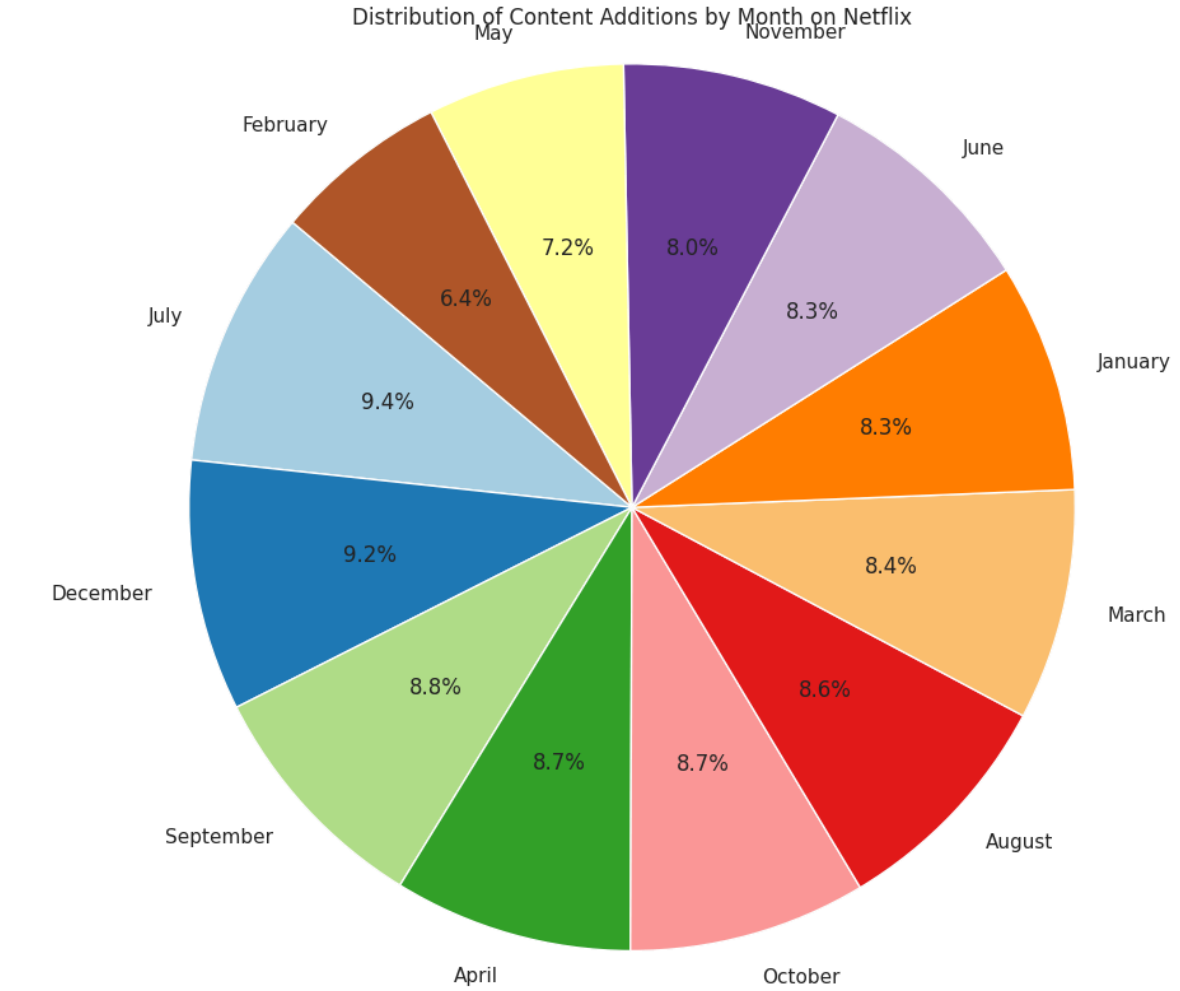
```
#Jagriti Tiwari
# Filter for movies only
movies_df = df[df['type'] == 'Movie'].copy()

# Extract duration in minutes, handling missing values and ensuring it's numeric
movies_df['duration_minutes'] = movies_df['duration'].str.extract(r'(\d+)\smin').astype(float)

# Group by release year and calculate the average duration
average_duration_by_year = movies_df.groupby('release_year')['duration_minutes'].mean().reset_index()

# Plotting the trend of movie durations over the years
plt.figure(figsize=(14, 7))
sns.lineplot(x='release_year', y='duration_minutes', data=average_duration_by_year, marker='o')
plt.title('Trend of Average Movie Duration on Netflix Over the Years')
plt.xlabel('Release Year')
plt.ylabel('Average Duration (minutes)')
plt.grid(True)
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

```
print(f"Trend of Average Movie Duration on Netflix Over the Years:\n{average_duration_by_year}")
```




```
Trend of Average Movie Duration on Netflix Over the Years:
    release_year  duration_minutes
0           1942         35.000000
1           1943         62.666667
2           1944         52.000000
3           1945         51.333333
4           1946         58.000000
..           ...               ...
68          2017         95.535248
69          2018         96.185137
70          2019         93.466035
71          2020         92.141199
72          2021         96.444043

[73 rows x 2 columns]
```

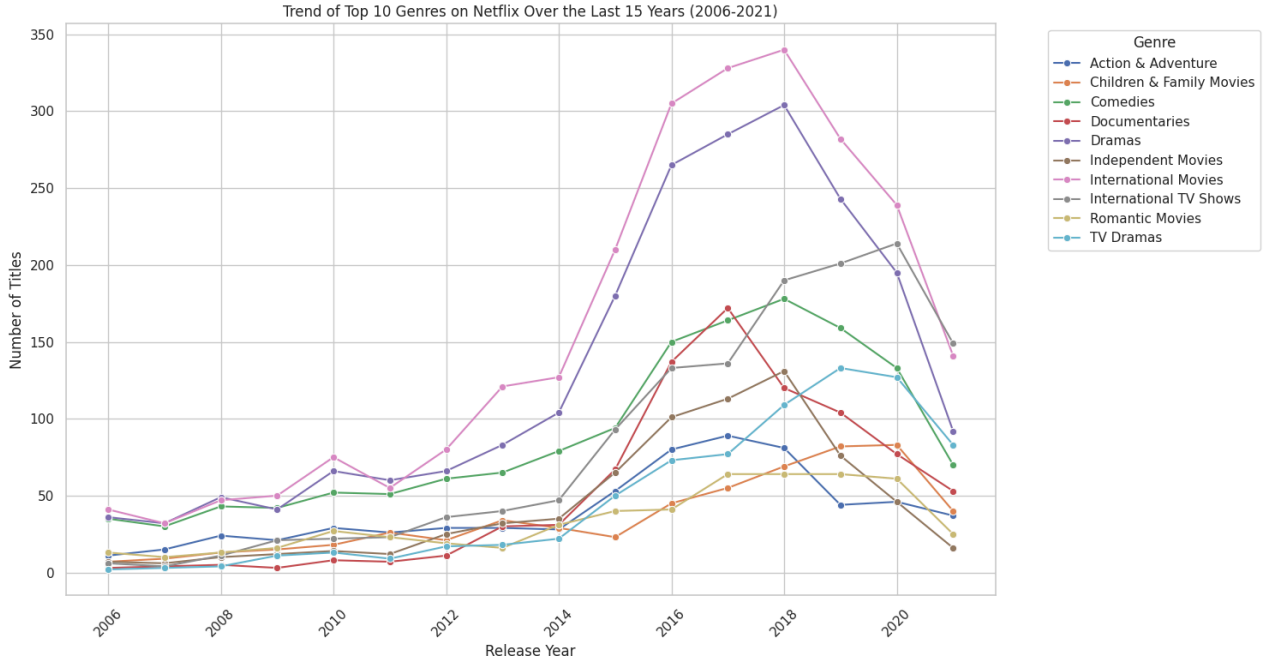
```
#Jagriti Tiwari

df['date_added'] = pd.to_datetime(df['date_added'], errors='coerce')
month_added_counts = df['date_added'].dt.month_name().value_counts()

plt.figure(figsize=(10, 8))
plt.pie(month_added_counts, labels=month_added_counts.index, autopct='%1.1f%%', startangle=140, colors=plt.cm.Paired.colors
plt.title('Distribution of Content Additions by Month on Netflix')
plt.axis('equal') # Equal aspect ratio ensures that pie is drawn as a circle.
plt.tight_layout()
plt.show()

print(f"Distribution of Content Additions by Month on Netflix:\n{month_added_counts}")
```




```
Distribution of Content Additions by Month on Netflix:
date_added
July         819
December     797
September    765
April        759
October      755
August       749
March        734
January      727
June         724
November     697
May          626
February     557
Name: count, dtype: int64
```

```
# Jagriti tiwari
# Prepare data: split genres and merge with release_year
genres_df = df.dropna(subset=['listed_in', 'release_year']).copy()
genres_exploded = genres_df.assign(listed_in=genres_df['listed_in'].str.split(', ')).explode('listed_in')
genres_exploded.rename(columns={'listed_in': 'genre'}, inplace=True)

# Count occurrences of each genre per year
genre_year_counts = genres_exploded.groupby(['release_year', 'genre']).size().reset_index(name='count')

# Get top N overall genres for plotting (e.g., top 10)
top_overall_genres = genre_year_counts.groupby('genre')['count'].sum().nlargest(10).index

# Filter data for top genres and recent years (e.g., last 15 years)
latest_year = df['release_year'].max()
years_to_plot = range(latest_year - 15, latest_year + 1) # Last 15 years

filtered_genre_year_counts = genre_year_counts[
    (genre_year_counts['genre'].isin(top_overall_genres)) &
    (genre_year_counts['release_year'].isin(years_to_plot))
]

# Plotting: Line plot for trend of top genres over years
plt.figure(figsize=(15, 8))
sns.lineplot(data=filtered_genre_year_counts, x='release_year', y='count', hue='genre', marker='o')
plt.title(f'Trend of Top 10 Genres on Netflix Over the Last 15 Years ({latest_year - 15}-{latest_year})')
```

```
plt.xlabel('Release Year')
plt.ylabel('Number of Titles')
plt.xticks(rotation=45)
plt.grid(True)
plt.legend(title='Genre', bbox_to_anchor=(1.05, 1), loc='upper left')
plt.tight_layout()
plt.show()

print(f"Trend of Top 10 Genres on Netflix Over the Last 15 Years ({latest_year - 15}-{latest_year})")
```




```
Trend of Top 10 Genres on Netflix Over the Last 15 Years (2006-2021)
```

```
# Jagriti Tiwari
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Drop rows with missing 'country' or 'listed_in' as these cannot form genre-country pairs
df_filtered_countries_genres = df.dropna(subset=['country', 'listed_in']).copy()

# Explode 'country' and 'listed_in' columns to get individual entries for each
df_countries_genres_exploded = df_filtered_countries_genres.assign(
    country=df_filtered_countries_genres['country'].str.split(', '),
    listed_in=df_filtered_countries_genres['listed_in'].str.split(', ')
).explode('country').explode('listed_in')

# Clean up any leading/trailing whitespace from country and genre names
df_countries_genres_exploded['country'] = df_countries_genres_exploded['country'].str.strip()
df_countries_genres_exploded['listed_in'] = df_countries_genres_exploded['listed_in'].str.strip()

# Rename 'listed_in' to 'genre' for clarity
df_countries_genres_exploded.rename(columns={'listed_in': 'genre'}, inplace=True)

# Calculate counts for each unique genre-country pair
genre_country_counts = df_countries_genres_exploded.groupby(['genre', 'country']).size().reset_index(name='count')

# Identify the top 10 overall genres based on total content count to focus the analysis
top_10_genres = genre_country_counts.groupby('genre')['count'].sum().nlargest(10).index

# Filter the genre-country counts to include only these top 10 genres
filtered_by_top_genres = genre_country_counts[genre_country_counts['genre'].isin(top_10_genres)]

# For each of these top 10 genres, find the top 5 countries that produce the most content
# This is done by grouping by genre and then selecting the top 5 entries within each group
top_countries_per_genre = filtered_by_top_genres.groupby('genre').apply(
    lambda x: x.nlargest(5, 'count')
).reset_index(drop=True)

# Create a FacetGrid plot to visualize the top countries for each of the top genres
# Each facet (sub-plot) will represent a genre, showing its top producing countries
```